Team Solutions

1. The other root of this quadratic equation is $3-2i$. A is -1 times the sum of the roots (6), B is the product of the roots (13), so the ordered pair is $\boxed{(-6, 13)}$.

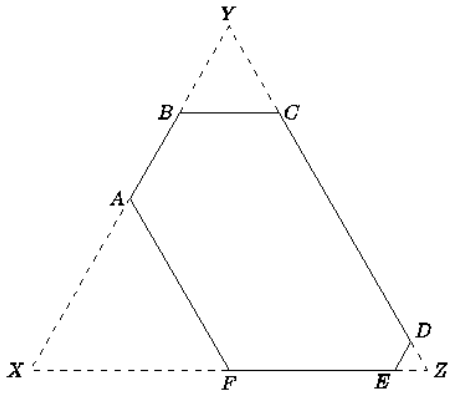
2. As the line is unspecified, let $f(x) = -3x$. $f(f(x)) = f(-3x) = -3(-3x) = 9x$ and $f(f(f(x))) = f(9x) = -3(9x) = -27x$. The sum of these three functions is $-21x$, so the slope is $\boxed{-21}$.

3. The expected amount of money earned per throw is \$18 (the average of the values on the faces), so the problem reduces to determining the expected number of throws. The game will end in 2 to 4 throws, with 2 throws occurring 1/3 of the time, 3 throws occurring 4/9 of the time (the probability the second throw is different than the first is 2/3, the probability the third throw is the same as either the first or second is 2/3), and 4 throws is $1-1/3-4/9 = 2/9$. Thus, the expected number of throws is $2\times(1/3)+3\times(4/9)+4\times(2/9) = 26/9$, so the expected winnings is $(26/9) \times \$18 = \boxed{\$52}$.

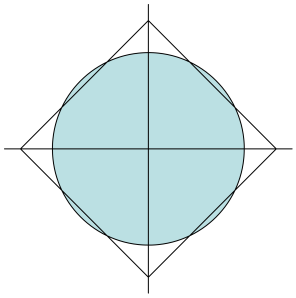
4. $3^{f(3)+f(9)} = 3^{\log_3 5+\log_9 11} = 3^{\log_3 5} \times 3^{\log_9 11} = 5 \times \left(9^{1/2}\right)^{\log_9 11} = 5 \times \left(9^{\log_9 11}\right)^{1/2} = \boxed{5\sqrt{11}}$.

5. Extend the sides $\overline{AB}$, $\overline{CD}$, and $\overline{EF}$ to obtain an equilateral triangle XYZ as shown. If we set $DE = u$ and $AF = v$, then we have $3+3+v = 3+8+u = u+5+v$, from which we conclude that $u = 1$ and $v = 6$. By subtraction of areas, we get:

$$[ABCDEF] = [XYZ] - [AFX] - [BYC] - [DZE] = \frac{\sqrt{3}}{4}\left(12^2 - 6^2 - 3^2 - 1^2\right) = \boxed{\frac{49\sqrt{3}}{2}}.$$



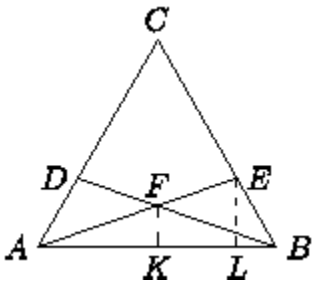
6. Points on the unit circle have coordinates $(\cos\theta, \sin\theta)$ for some value of θ, $0 \leq \theta \leq 2\pi$. So, we can view these points on the plane. $|x|+|y| = k$ describes a square with corners at $(\pm k, 0)$ and $(0, \pm k)$. Each edge of this square intersects the unit circle twice, provided $1 < k < \sqrt{2}$, so there are $\boxed{8}$ points of intersection.



7. There are $\binom{5+5-1}{5-1} = 126$ arrangements of the red balls, all equally likely. You win if and only if your opponent puts three or more red balls in any single box. For example: [3,1,1,0,0] means that your opponent wins the first box and you win the last two. There are five arrangements with five balls in one box, twenty arrangements with four in one box and a single ball in the other, and fifty arrangements with three in one box, and either two in one other box or a single ball in two boxes. The probability of winning is $\frac{50+20+5}{126} = \frac{75}{126} = \boxed{\frac{25}{42}}$.

8. The coefficient of the term $x^a y^b z^c$ is $\frac{2009!}{a!b!c!}$, where we assume $a+b+c = 2009$. The number of powers of five in the numerator is $\left\lfloor \frac{2009}{5} \right\rfloor + \left\lfloor \frac{2009}{5^2} \right\rfloor + \left\lfloor \frac{2009}{5^3} \right\rfloor + \left\lfloor \frac{2009}{5^4} \right\rfloor = 401 + 80 + 16 + 3$. In order for this to equal the number of powers of five in the denominator, we need $\left\lfloor \frac{a}{5^k} \right\rfloor + \left\lfloor \frac{b}{5^k} \right\rfloor + \left\lfloor \frac{c}{5^k} \right\rfloor = \left\lfloor \frac{2009}{5^k} \right\rfloor$ for all k. Note that the floor function is sub-additive, so we will always have $LHS \leq RHS$. Noting that $2009_{10} = 31014_5$, we first distribute the three powers of 5^4 among a, b, and c. There are $\binom{3+3-1}{3-1} = 10$ ways to place them. After doing this, we have placed 15 of the 16 powers of 5^3, so we only get to distribute one more, and there are $\binom{1+3-1}{3-1} = 3$ places for it to go. After this, all 80 powers of 5^2 have been taken care of, and we have 1 leftover power of 5 to place (3 places for it to go as well). After this, we have coefficients summing to 2005, so we can place the remaining 4 powers anywhere: there are $\binom{4+3-1}{3-1} = 15$ places for these. Overall, there are $10 \times 3 \times 1 \times 3 \times 15 = \boxed{1350}$ terms whose coefficients are not multiples of 5.

9. The area equals $\frac{AD \cdot AF \cdot \sin \angle DAF}{2} = \frac{AF \cdot \sin \angle CAE}{2}$. Dropping perpendiculars from E and F to $\overline{AB}$ at points K and L we get $BL = \frac{1}{2}$ and $\frac{AF}{AE} = \frac{AK}{AL} = \frac{3/2}{5/2} = \frac{3}{5}$. The area of triangle ADF is: $\frac{3}{10} \cdot AE \cdot \sin \angle CAE = \frac{3}{10} \cdot CE \cdot \sin \angle ACE = \frac{3}{10} \cdot 2 \cdot \frac{\sqrt{3}}{2} = \boxed{\frac{3\sqrt{3}}{10}}$ using the Law of Sines on $\triangle ACE$.



10. The answer is $\boxed{2, 1, 4, 2, 2}$, and the solved puzzle is shown below. There are many ways to get to the final solution. The key thing to note is the unique arrangements of digits in the L-shaped $75\times$, $16\times$, and $2\times$ cages immediately give you the lower half of the puzzle, since the $1-$ cage in the lower right corner is uniquely determined by the other digits in the L-shaped cages.

